A function $f : A \to B$ is called an into function if there exists atleast one element in B which is not the image of any element of A.

That is the range of f is a proper subset of the co-domain of f.

In other words, a function $f : A \to B$ is called 'into' if it is not 'onto'.

Illustration 13

Let $A = \{1,2,3,\}$ and $B = \{w, x, y, z\}$, $f = \{(1, w), (2, z), (3, x)\}$

Here, range of $f = \{w,\ x,\ z\} \subset B$ (Fig.1.28)

$\therefore$ f is a into function.

Note that $y \in B$ is not an image of any element in A.

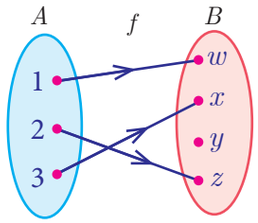


Fig. 1.28

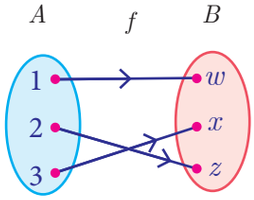
1.7.5 Bijection

Consider the circle where each letter of the English alphabet is changed from inner portion to a letter in the outer portion. Thus $A \to D$, $B \to E$, $C \to F$, ... $Z \to C$. We call this circle as 'cipher circle'. (Fig.1.29) In this way if we try to change the word 'HELLO' then it will become 'KHOOR'. Now using the same circle if we substitute for each outer letter the corresponding inner letter we will get back the word 'HELLO'. This process of converting from one form to an other form and receiving back the required information is called **bijection**. This process is widely used in the study of secret codes called **cryptography**.


Cipher Circle
Fig. 1.29

If a function $f : A \to B$ is both one–one and onto, then f is called a bijection from A to B.

Illustration 14

one to one and onto function (Bijection)	
 Fig. 1.30	Distinct elements of A have distinct images in B and every element in B has a pre-image in A.

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